

Suppressing the Loss of Polymer-based Dielectrics for High Power Energy Storage

Jia-Yao Pei, Li-Juan Yin, Shao-Long Zhong, Zhi-Min Dang*

State Key Laboratory of Power System, Department of Electrical Engineering, Tsinghua University, Beijing
100084, P. R. China

E-mail: dangzm@tsinghua.edu.cn

Abstract

Polymer-based dielectrics have received intensive interest from academic community in the field of high-power energy storage owing to their superior flexibility and fast charge-discharge ability. Recently, how to suppress the loss of polymer-based dielectrics has been increasingly recognized as a critical point to attain a high charge-discharge efficiency in the film capacitors. Some achievements have been made in analyzing the source of loss and suppressing loss via Edison's trial and error method. In this review, the significance of suppressing loss in polymer-based dielectrics is firstly emphasized. Then, different sources of loss are discussed carefully and an in-depth analysis of the related measurements is presented. Next, recent research results in suppressing loss are summarized and discussed in detail according to different strategies. Lastly, the challenges and opportunities in the loss suppression research for the rational design of high-efficiency polymer-based dielectrics are proposed.

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the [Version of Record](https://onlinelibrary.wiley.com/doi/10.1002/adma.202203623). Please cite this article as [doi: 10.1002/adma.202203623](https://onlinelibrary.wiley.com/doi/10.1002/adma.202203623).

This article is protected by copyright. All rights reserved.

1. Introduction

Nowadays, electric energy is essential for industry and daily life due to its easy transmission, efficient conversion, cleanliness, and eco-friendliness^[1]. Energy storage is indispensable to improving the efficiency and stability of electric energy utilization^[2]. Compared with other energy storage devices including supercapacitors and batteries, capacitors feature higher power density due to their ultrafast charge-discharge characteristics^[3]. Capacitors with polymer films as dielectric layers comprise ~50 % of the global high voltage capacitor market due to their high breakdown strength, easy processing, and favorable self-healing performances^[4].

The discharge energy density U_e of the dielectric polymer materials can be calculated as $U_e = \int_{D_{\max}}^{D_{\text{rem}}} E dD$, where E is the applied electric field, $D_{\max}$ is the maximum electric displacement and D_{rem} is the remanent electric displacement. For linear dielectric materials, the equation can be expressed as $U_e = \frac{1}{2} \varepsilon_0 \varepsilon_r E^2$, where ε_0 is the vacuum dielectric constant and ε_r is the relative permittivity of the material. Nevertheless, the relative permittivity of biaxially oriented polypropylene (BOPP), the industrial benchmark polymer dielectric, is rather low ($\varepsilon_r \sim 2.2$), resulting in a limited energy density and an excessive device size^[5]. Substantial progress has been made in enhancing the energy density of polymer-based dielectric materials in this field^[6]. However, reported dielectric materials with improved energy density are often accompanied by elevated dielectric loss and deteriorated efficiency^[7]. These materials still fail to meet industrial requirements, as their efficiency is usually much lower than that of BOPP^[8].

The loss of dielectrics is one of the key factors that determines whether dielectric materials can be used in film capacitors. Recently, dielectric loss has drawn growing research interest, as its significance in academic research and industrial application has been gradually noticed^[9]. The dielectric loss not only sets an upper limit on the efficiency of film capacitors, but also generates heat, posing severe challenges

to the heat management of the whole electric set^[10]. In addition, elevated temperature in capacitors can lead to further deterioration of efficiency, greatly shortened lifetime, and even thermal runaway in the worst case^[11].

Some progress has been made in suppressing the loss of dielectrics^[12]. Yet, these loss suppression strategies may differ greatly in method and mechanism due to the differences in the numerous polymer matrices, functional fillers and routes employed in different works. This poses difficulties for analyzing and understanding the loss suppression of polymer-based dielectric. Currently, there is no review on the sources of different loss types and the classification of research achievements in this field, which impedes further development of polymer-based dielectrics with high efficiency. Reviewing the research progress of loss suppression is vital for researchers in this field to get the full picture and find the crux of the matter.

This review starts with the sources and classifications of dielectric loss, which is helpful to understand how different loss suppression methods work and what are the main sources of loss in different situations. Second, an in-depth analysis of the most commonly used measurements is presented. The effect of different loss types on the measurement results is analyzed in this section. Then, a detailed and well-classified review of recent progress in suppressing the loss of polymer-based dielectric is conducted. Lastly, the challenges and opportunities in loss suppression for the rational design of high-efficiency polymer-based dielectrics are addressed, which may be enlightening for future research.

2. Source and Measurement of Dielectric Loss

There are two main types of dielectric loss: conduction loss and polarization loss, which originate from the behaviors of free charges (also known as carriers) and bound charges, respectively^[13]. The conduction loss produced by collision, trapping and recombination during the movement of free charges is unavoidable in charged capacitors^[13-14]. The energy loss resulting from the relative displacement of bound charges with different polarities during polarization and depolarization is called polarization loss^[13]. In the following, the source of conduction loss and polarization loss is introduced, and corresponding methods of loss measurement are also analyzed in detail.

2.1 Conduction Loss

Polymers used for insulation usually feature a wide bandgap and rather low carrier concentrations. However, under high applied electric fields and elevated temperatures, it becomes much easier for the charges to be excited as carriers or injected from the electrodes^[14]. Therefore, the conductivity of dielectric usually rises significantly under high electric fields, resulting in severe energy loss.

The most common conduction mechanisms for dielectric polymer films can be divided into two types: electrode-limited conduction mechanisms (Richardson-Schottky emission, Fowler-Nordheim tunneling) and bulk-limited conduction mechanisms (Ohmic conduction, Poole-Frenkel emission, Hopping conduction, Space-charge-limited conduction)^[15].

Richardson-Schottky emission assumes that electrons could obtain energy from thermal activation, overcome the energy barrier at the electrode-dielectric interface, and enter the dielectric. Richardson-Schottky emission can be expressed by^[16]

$$J = AT^2 * \exp\left[-\left(\frac{\varphi - \sqrt{q^3/4\pi\epsilon\sqrt{E}}}{kT}\right)\right] \quad (1)$$

where J is the current density, E is the electric field, A is the Richardson constant, φ is the barrier height at the electrode-dielectric interface, q is the charge of the carrier, T is the temperature, ϵ is the dielectric constant of dielectric material, and k is the Boltzmann constant. For example, the leakage current type of polyethylene-terephthalate and polyvinylformal displays the Richardson-Schottky characteristics at electric fields between 2 kV/mm and 20 kV/mm and temperatures from 25°C to 100°C.^[17]

Fowler-Nordheim tunneling can be observed at a relatively high applied electric field. A considerable amount of charges can be found to cross the barrier and exist in the bulk of dielectric under a high electric field due to quantum mechanics. The expression of Fowler-Nordheim tunneling^[16, 18] is

$$J = \frac{q^3 E^2}{8\pi h \varphi} * \exp\left[\frac{-8\pi(2qm_T^*)^{1/2}}{3hE} \phi_B^{3/2}\right] \quad (2)$$

where h is the Planck's constant, $\phi_B = \frac{\varphi}{q}$, m_T^* is the tunneling effective mass and the other notations are the same as defined before. Zhang and co-workers^[19] revealed that the dramatic decrease of charge-discharge efficiency of several low loss dielectric materials at high electric fields was attributed to the increased conduction loss featuring Fowler-Nordheim tunneling.

The current density of ohmic conduction is given as

$$J = E/\rho \quad (3)$$

where ρ is the volume resistivity and the other notations are the same as defined before. The ohmic conduction is usually the dominant current under a low electric fields and ρ is affected by temperatures.

Poole-Frenkele emission describes that the trapped electrons obtain energy from thermal excitation and become carriers in the bulk. The current of Poole-Frenkele emission is expressed as^[18]

$$J = q\mu N_C E * \exp\left[-\frac{(\varphi_B - \beta_{PF}\sqrt{E})}{kT}\right] \quad (4)$$

where μ is the electronic drift mobility, N_C is the density of states in the conduction band, φ_B is the trap energy level, β_{PF} is the constant of Poole-Frenkel, and the other notations are the same as defined before. Hadri et al^[15c] found that the field threshold (E_T), which is defined as the switch from an ohmic to a strongly non-linear behavior, was within the range of 150 kV/mm and 160kV/mm for semi crystalline polyethylene terephthalate. The mechanism of conduction is identified as the Poole-Frenkel mechanism in higher electric fields.

Hopping conduction assumes that the electron obtains energy and ‘hop’ from one localized state to another^[20]. The current density J is given as^[15b, 15c]

$$J = 2nqv\lambda * \exp\left(-\frac{W_a}{kT}\right)\sinh\left(\frac{\lambda qE}{2kT}\right) \quad (5)$$

where n is the carrier concentration, v is the attempt-to-escape frequency, λ is the hopping distance, W_a is the activation energy, and the other notations are the same as defined before. Janet ho et al^[21] found that conduction of BOPP followed the hopping mechanism and the thermal activation energy was about 0.76 eV. Besides, the fitting of leakage current of polyimide (PI), polyetherimide (PEI) and cross-linked divinyltetramethyldisiloxane-bis(benzocyclobutene) (c-BCB) suggested that hopping conduction was dominant at high electric fields (usually between 50 and 300 kV/mm) and elevated temperatures^[22].

Space-charge-limited conduction is given by Child’s law^[16, 23]

$$J = \frac{9}{8}\varepsilon\mu\theta\frac{V^2}{d^3} \quad (6)$$

where ε is the dielectric constant, μ is the carrier mobility, θ is the ratio of the free carrier density to total carrier (free and trapped) density, V is the applied voltage and d is the thickness of the dielectric film. Montanari et al^[24] found that the threshold electric fields for space charge accumulation were usually between 5 and 20 kV/mm for several kinds of polyethylene and polypropylene, which indicated that space-charge-limited conduction was the most likely mechanism for these polymers

The parameters of conduction mechanisms indicate feasible ways for the suppression of conduction loss. For example, the Richardson-Schottky emission is strongly related to the electric field, permittivity and barrier height at the electrode-dielectric interface. As been demonstrated in previous research results, suppressing the electric field, increasing the permittivity and reviving the barrier height at the electrode-dielectric interface all show great contributions to the enhancement of breakdown strength and discharged energy density^[25]. Besides, the conduction mechanism of polyetherimide (PEI) at elevated

This article is protected by copyright. All rights reserved.

temperature and high electric fields is identified as the hopping conduction^[22b]. It has been experimentally proved that the introduction of the deep traps decreases the hopping distance and reduces the conduction loss at elevated temperatures^[22a].

Nevertheless, the identification of current types does not necessarily mean that there is only one way to suppress the conduction loss. For example, the conduction mechanism of PEI at elevated temperature and high electric fields is identified as the hopping conduction^[22b], which belongs to the type of bulk-limited current. Yet, the nano interlayer between the dielectric and electrode has been verified to be effective in reducing the charge injection and improving the discharged efficiency^[26].

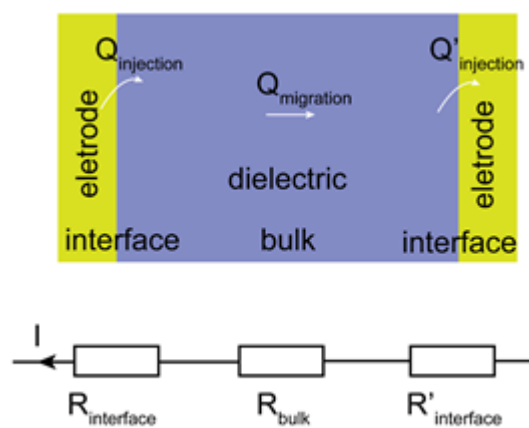


Figure 1. Schematic of the charge injection from the electrodes and charge migration in the bulk, and the model of series resistors.

As demonstrated in Fig. 1, when the current type is defined as one type of limited current (interface or bulk), it means that the charge transport at the other place is no longer the 'limited' factor and becomes more severe. Therefore, the current types also give us another implicit hint to solve the loss problem. The charge transport at the interface and in the bulk are all noteworthy in the inhibition of leakage current.

2.2 Polarization Loss

There are six types of polarization, including electronic, atomic, dipolar, ionic, spontaneous space and charge polarization in the polymer dielectrics when applied with an electric field (Fig. 2)^[4c, 9a, 13, 27].

Electronic polarization and atomic polarization can be able to respond to extremely high frequency ($>10^{10}$ Hz) and the loss peak can be found in the infrared and optical range of frequency. Hence, loss originating from these polarizations usually can be ignored in polymer dielectrics as the operation frequency of most capacitors is commonly less than 1 GHz.

This article is protected by copyright. All rights reserved.

Ionic species usually introduced from polymer production, which leads to the ionic polarization in the dielectric. When ion species are confined in the local area, it can be regarded as a kind of atomic polarization. However, the migration of ions in the polymer matrix under high electric fields can cause a huge increase in dielectric loss and electrical conduction. A recent report^[28] has shown that confined ions could serve as deep traps and improve the breakdown. Still, the leakage current of polymer dielectrics in this work increased with the introduction of ions.

The relax frequency of dipole polarization can be between 10 Hz and 10^{10} Hz^[9a], which may reside within the operating frequency of capacitors. Therefore, the dipolar polarization loss should be considered when designing the polymer-based dielectrics for high power energy storage.

The spontaneous polarization occurs in the materials with non-centrosymmetric crystalline structure, where the centroid of the negative charges does not coincide with that of the positive charges. The orientation of the polar domain and the unreleased energy due to the spontaneous polarization can cause serious energy loss in ferroelectric polymer dielectrics. Polyvinylidene fluoride (PVDF) and its copolymers are some of the most commonly used polymer matrices in this field due to their high permittivity and high breakdown strength. The crystallization of δ , γ , and β form in PVDF and its copolymers are ferroelectric, which induces the spontaneous polarization in the polymer dielectrics.

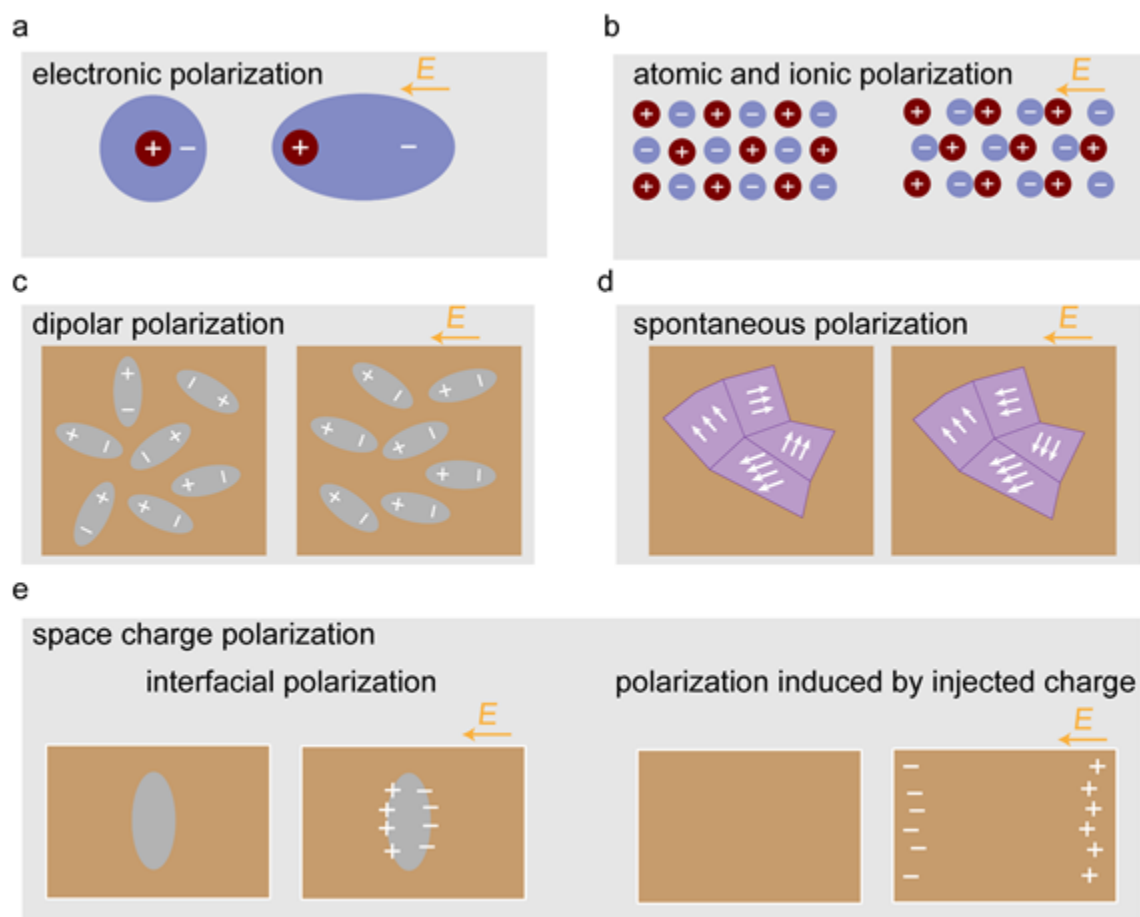


Figure 2. Schematic of a) electronic polarization, b) atom and ionic polarization, Reproduced with permission^[27a]. Copyright 2017 Elsevier. c) dipolar polarization, Reproduced with permission^[4c]. Copyright 2021 Royal Society of Chemistry. d) spontaneous polarization Reproduced with permission^[27b]. Copyright 2016 American Chemical Society. and e) space charge polarization.,

Space charge polarization can be found in the polymer matrix containing mobile and trapped charges. Moreover, the activated carriers and the charges injected from the electrode can further contribute to the space charge polarization and corresponding loss.

Therefore, the losses of dipolar polarization, spontaneous polarization and space charge polarization are the most important dielectric loss types for capacitor dielectric films. A detailed analysis of these polarization losses will be given below.

2.2.1 Dipole Polarization Loss

Dipole polarization loss is a great contributor to the polarization loss in linear dielectrics, which originates from the 'friction' within the polymer during the orientation of the dipoles under electric fields. Therefore, the effective method for suppressing the dipole polarization loss is to reduce the 'friction' within the polymer during the orientation. Zhu' group^[29] prepared novel sulfonylated polymers(SO₂-PIM)

This article is protected by copyright. All rights reserved.

of intrinsic microporosity (0.5~1.25 nm), which provided abundant free volumes for friction-free rotation of side-chain dipolar groups. As a consequence, this polymer shows a dissipation factor as low as 0.005, while achieving a high dielectric constant ($\epsilon_r = 6$).

Besides, As the temperature rises and approaches the glass transition temperature (T_g), the motions of chains segments become easier, resulting in the increase of the dielectric loss. When the ambient temperature is higher than T_g , the chaotic thermal motions of the dipoles give an obstacle to their orientations toward the external electric field, leading to a decrease in the dielectric loss^[30]. The α relaxation is dominant in the vicinity of glass transition and the DC conductivity also increases remarkably^[31]. Zhu et al^[8b, 32] have shown that a dipolar glass polymer with a high T_g (suppressing α relaxation) and a significant dipolar polarization in sub- T_g transitions (β and γ transition for high permittivity) shall be a promising candidate for high energy density, high temperature, and low loss dielectrics.

2.2.2 Spontaneous Polarization Loss

The bulky and strongly interactive ferroelectric domains are thought to induce serious energy loss during the polarization and store vast amounts of unreleased energy even after the electric field is removed^[33]. Thus, it has been proved to be effective in suppressing energy loss by reducing ferroelectric crystallization, decreasing the ferroelectric domain size and inhibiting the interaction between different domains^[9a]. Research progress has been made in enhancing the energy density of ferroelectric polymer-based dielectrics. Yet, the efficiency of ferroelectric polymer-based dielectrics is usually much-below 90%, which greatly hinders their applications.

2.2.3 Space Charge Polarization Loss

There is space charge accumulation in the interface of different parts in an inhomogeneous dielectric with mismatched permittivity and conductivity (Maxwell-Wagner effect). The redistribution of charges in the interface enhances the polarization of dielectrics, which is called interfacial polarization and can be considered as a kind of space charge polarization.

The interfacial polarization and the polarization in the interphase are sometimes inappropriately confused. Due to the diverse permittivity of different phases in the dielectric polymer composites, severe electric field distortion can be found in the interphase and these phases have a higher energy density compared with the average energy density of the polymer composite^[34]. The interfacial polarization can contribute to the polarization in the interphase due to the electric field redistribution with the introduction of the space charge. Still, they are not the same, because in a hypothetical multiphase insulator with no space charge, the enhanced polarization in the interphase due to the discrepancy of

This article is protected by copyright. All rights reserved.

permittivity still exists.

Moreover, another kind of space charge polarization can be induced at high electric fields. The charges injected from the electrodes are not equal to the charges transferred in the matrix, which leads to charge accumulation near the electrode-dielectric interface. This can also contribute to the space charge polarization and induce energy loss. Up to now, this phenomenon has received little attention and will be discussed in the following part of 2.3.2.

2.3 Method of Loss Measurement

Currently, the loss tangent, the area enclosed in the electric displacement-electric field (DE) loop (Fig. 3a) and leakage current are three main representative parameters of the dielectric loss.

2.3.1 The loss tangent

The loss tangent represents the sum of conduction loss and polarization loss at a given electric field and frequency. There are several theoretical and empirical models, which can be used to analyze the dielectric response, including Debye model, Cole-Cole model, Davidson-Cole model Havriliak-Negami model and Dissado-Hill model^[35]. The analysis of the loss tangent of the dielectrics with these models can clarify the contribution of different losses and establish the relationship between the loss and material structures^[36].

The loss tangent is usually measured over a wide frequency band (1mHz~10 MHz) but at low voltages (lower than 5 V) which is far below the working/operating voltage (usually >100 V) for polymer-based dielectrics. However, the conduction loss of dielectric rises sharply with the increase of the electric field. Also, the loss of spontaneous polarization and space charge polarization is strongly related to the electric field. Therefore, the loss tangent at low electric field can not fully demonstrate the loss of polymer-based dielectrics operating at high electric field.

2.3.2 Area enclosed in the DE loop

DE loop is another common way to demonstrate the discharged energy and efficiency of dielectric materials. The DE loop is usually obtained by applying a triangle wave of 10 Hz or 100 Hz to the dielectric materials. After recording the voltage and current or charge in the measurement, the D can be calculated by the measured current or charge and the E can be obtained by the voltage. Fig. 3a shows a typical DE loop of polymer dielectrics. The loss is also a sum of polarization loss and conduction loss.

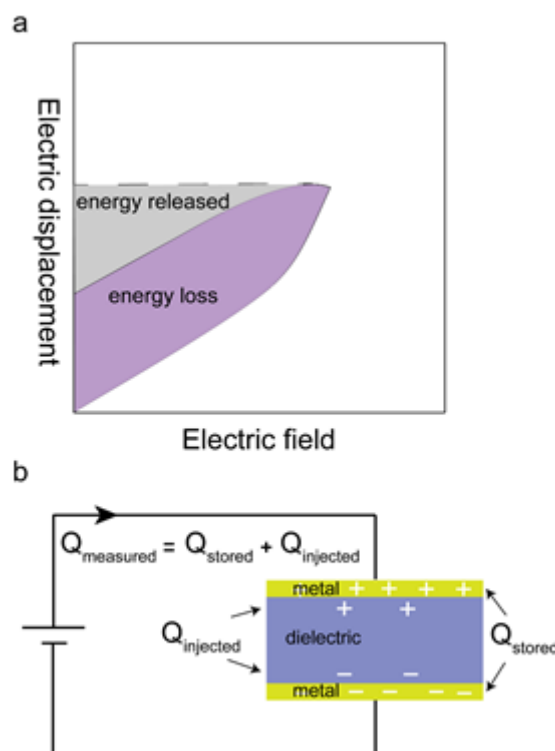


Figure 3. Schematic of a) DE loop and b) measured charge, which can be divided into two parts: stored charge on the electrodes and injected charge into the dielectrics.

As shown in the Fig. 3b, the measured electric displacement is the sum of the contributions of the charges injected into the dielectric and the charges stored on the electrode. The injected charges into the dielectric could serve either for leakage current or become space charge.

However, the leakage current at elevated temperatures and high fields needs a long time to stabilize, due to the balancing process of injected charges and charge migration in the bulk. Besides, the measurement process of DE is not the same as the discharge process of capacitors. Therefore, the contribution of different loss types to dielectric loss may differ considerably under various test conditions. Further research is still needed about this measurement and more details of the loss can be extracted from it.

2.3.3 Leakage current

Different from the loss tangent and DE loop, the leakage current mainly represents the conduction loss of dielectrics. However, the measurements of leakage current could be affected by many factors and great caution is required to obtain credible results. For example, electric fields can influence the crystallization of PVDF and its copolymers^[9], and further affect the space charge and leakage current in the dielectrics. These processes complicate the measurement of leakage current and weaken the

repeatability of measured results.

It is worth noting that the leakage current in some literature first increases and then decreases with the increases of electric field^[37]. It deserves more research attention, as all current mechanisms show a monotonical increase with the increase of electric field intensity. This interesting phenomenon can be attributed to two reasons. First, the time to reach stability increases with the voltage rising and the current may not fully stabilize. Second, there are crystal transformation and crystal switching in the PVDF matrix with the increase of electric field, which may lead to the change of resistance and capacitance and influence the leakage current. Further investigations on this phenomenon would provide more information for PVDF.

Furthermore, it is quite confusing that the measured values of leakage current vary greatly, even by orders of magnitude. For instance, a wide variation in the leakage current of pristine PEI can be found in different literature^[25d, 38]. Besides, some leakage currents at lower temperatures and electric fields show higher values compared with that at higher temperatures and electric fields^[22b, 25d]. This may be attributed to the preparation process of PEI. Still, the measurement of leakage current should be carefully checked for reliable results.

In summary, the measurement of dielectric loss is far from satisfactory. On the one hand, the measurements of different types of loss and the characterizations of the charge transportation need be further developed to identify the ratio of different types of loss in the dielectrics. For example, there is still a lack of means to measure the internal charge distribution of thin materials at high electric fields. The resolution of commercially available Pulsed Electro-Acoustic (PEA) system is usually about 10 μm , which is close to the thickness of common dielectric polymer films. Currently, PEA measurements of thick films (usually $>100\mu\text{m}$) of the same material are used to study the charge dynamics of thin dielectric polymer films. The electric field of PEA tends to be much smaller than that of breakdown strength due to the decrease of dielectric strength resulting from the increased film thickness. Therefore, the measurement of space charge distribution in the thick samples cannot fully represent the charge distribution in thin films. The inner process of charge transportation is speculated by the measurement of leakage current in most present research works.

On the other hand, the dielectric loss for different capacitor applications needs more investigations and understanding. The leakage current at operation electric fields and the loss caused by ripple voltage should be considered for capacitor serving for DC filtering. However, for capacitor served for fast charge and discharge, the measurement of the loss during the fast charge and discharge process should be stressed. Different applications have different requirements for each type of loss, which has received much less attention. Therefore, it is suggested that the measurement of loss should be further developed

This article is protected by copyright. All rights reserved.

for the research and application of polymer-based dielectrics for high power energy storage.

3. Design of Polymer-based Dielectric Materials with Suppressed Loss

3.1 Nonferroelectric materials

The main methods to suppress the loss of nonferroelectric materials can be categorized into two types: reducing the polarization loss and suppressing the conduction loss.

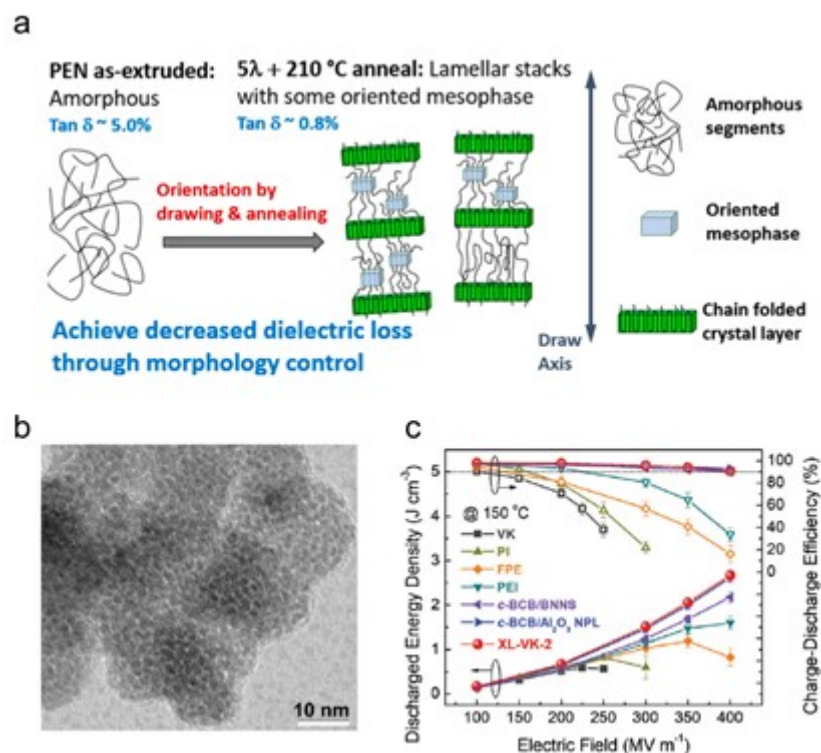


Figure 4. a) Morphology progression of poly (ethylene naphthalate) films. Reproduced with permission^[39]. Copyright 2017 American Chemical Society. b) Bright-field TEM micrograph of SO₂-PIM. Reproduced with permission^[29]. Copyright 2014 Royal Society of Chemistry. c) Discharged energy density and charge-discharge efficiency of high-temperature dielectrics at 150 °C, respectively. Reproduced with permission^[40]. Copyright 2020 Royal Society of Chemistry.

Reducing the polarization loss of the nonferroelectric materials mainly focuses on reducing the 'friction' of dipole orientation. Miranda et al^[39] used polyethylene naphthalate (PEN) to study how morphological factors inhibited chain segment relaxations. They found that film orientation greatly reduced loss through strain-induced crystallization and the oriented amorphous mesophasic regions (Fig. 4a). Zhu's group^[29] prepared novel sulfonylated polymers (SO₂-PIM) of intrinsic microporosity (0.5~1.25 nm), which provided a large free volume for friction-free rotation of side-chain dipolar groups (Fig. 4b). This polymer showed a dissipation factor as low as 0.005, while a high dielectric constant ($\epsilon_r = 6$). Zhang's group^[41] found that low nanofiller loading of inorganic fillers (Al₂O₃, lower than 1 vol%) in the

polyarylene ether urea (PEEU) matrix could greatly enhance the permittivity of dielectric composites and suppress the loss tangent within a wide temperature range. The enhanced performance was attributed to the reduced constraints on dipole motions in the glassy state of polymers.

Recently, suppressing conduction loss has received increasing attention. Strategies including polymer design, crosslinking and grafting are found to be effective to reduce the conduction loss at elevated temperatures. A kind of polyolefin (polyoxafluoronorbornene, POFNB) consisting of repeating units of fairly rigid fused bicyclic structures and alkenes separated by freely rotating single bonds was reported^[42]. POFNB with a wide bandgap (≈ 5 eV) exhibited an electrical conductivity two orders of magnitude lower than the best commercial high-temperature polymers. In addition, crosslinked fluoropolymers (XL-VK-2) exhibited improved charge-discharge efficiency (Fig. 4c)^[40]. The marked improvement in efficiency originated from efficient charge-trapping by a range of the molecular trapping centers, which were generated from the crosslinked structures. Meanwhile, crosslinking could limit the movement of polymer chains at high temperatures and high applied electric fields, which improved the temperature resistance of polymer materials and reduced dielectric loss^[43]. The band structures of the grafted polymer could introduce deep traps compared with the pristine polymer. The deep traps would capture carriers and inhibit the charge transport, which led to decreased conduction current and suppressed space charge injection and accumulation^[44]. Zhang et al^[45] reported a series of self-cross-linkable oligomers prepared by combining the advantages of PEI and phenylethynyl groups. After crosslinked under oxygen, dielectric films with enhanced charge-discharge efficiency at elevated temperatures and low leakage current density were obtained.

To sum up, the polarization loss of the nonferroelectric materials demands more experimental and theoretical efforts. Systematic investigations need be devoted in the process technologies, which can effectively regulate the polymer morphology to suppress the polarization loss. In addition, the composite dielectrics with ultra-low loading of inorganic fillers could achieve much-improved permittivity and reduced loss^[46]. This phenomenon needs to be carefully studied to know how the interaction between the fillers and the matrix works, as ultra-low loading of inorganic fillers may be favorable for film scale-up processes. Besides, deep traps have been popular in analyzing the performance of polymer-based dielectrics. Yet, the role of deep traps is empirical and few quantitative descriptions of the effect of deep traps on the conduction and insulation of dielectrics have been reported. Last but not least, the rational design of polymer dielectrics via the data-driven and computation methods^[47] shows great potential in designing polymer dielectrics with wide bandgap and high permittivity. Research advances in this field can greatly accelerate the development of new materials^[48].

3.2 Ferroelectric Polymers

This article is protected by copyright. All rights reserved.

PVDF and its copolymers featuring high permittivity and breakdown strength suffer from serious energy loss due to spontaneous ferroelectric loss. The crystal orientation, crystallite size and crystallinity all have effects on the loss of ferroelectric polymers^[9a, 49]. Processing technology and chemical modification are mainly utilized to reduce spontaneous polarization loss.

Parallel PVDF crystal orientation may work better at elevated temperatures for the low loss purpose, and decreasing the overall crystallinity is beneficial for reducing the hysteresis loops^[9a]. Meng et al^[50] adopted the pressing-and-folding approach (Fig. 5a) to prepare PVDF with ultrahigh β -phase (98% of crystalline phase, decreasing crystallite size ~ 4 nm). The prepared dielectric films showed relaxor-like behavior and delivered an ultra-high energy density (35 J cm^{-3}) with high efficiency. Yuan et al^[33d] explored a series of stretched poly(vinylidene fluoride-co-hexafluoropropylene) (P(VDF-HFP)). They found that increased crystal orientation and reduced crystallite size facilitated dipole orientation, and recoverable energy density was improved due to the formation of reversible ferroelectric nanodomains accordingly. Guo et al^[51] investigated the effect of stretch ratio on the stretched PVDF films by uniaxial stretching process. The phase transition and crystal orientation during mechanical stretching process led to improved energy efficiency at high electric fields. Jiang et al^[52] compared nonequilibrium nanocomposites (including electrospinning, hot-pressing and thermal quenching) with equilibrium nanocomposites (via solution casting). Favorable phase composition and small crystalline size were induced by the nonequilibrium process, which was beneficial for the reduction of remnant polarization and the enhancement of efficiency at high electric fields (Fig. 5b).

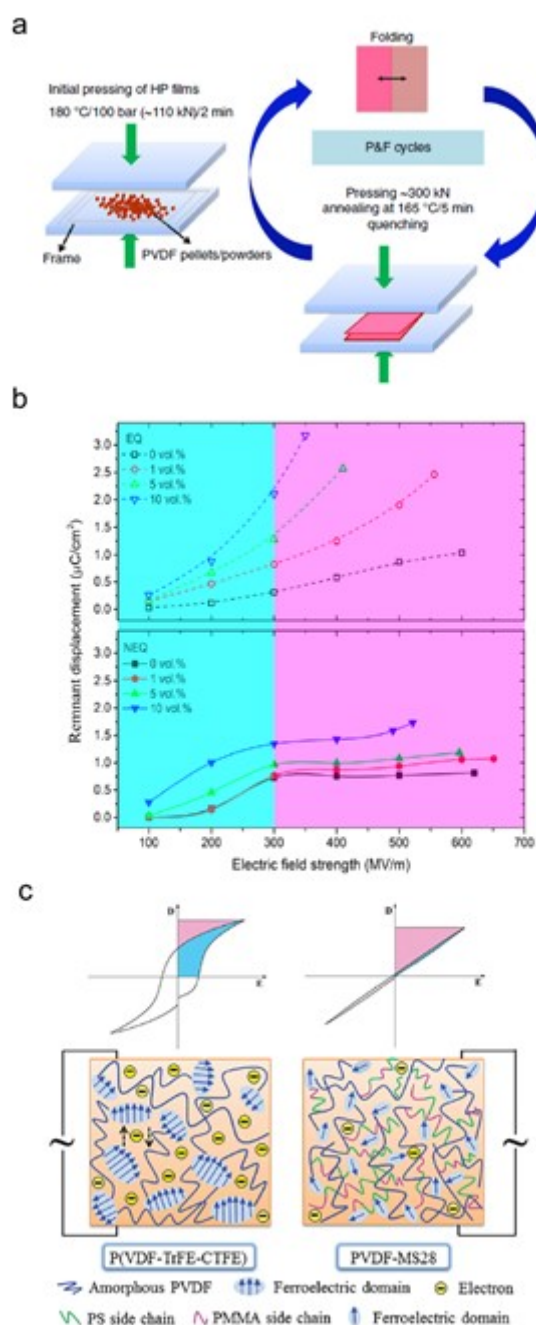


Figure 5. a) Schematic of pressing and folding process^[50]. Reproduced from ref. 50 with open access from Springer Nature. b) Remnant displacement of the equilibrium (top) and nonequilibrium (bottom) nanocomposite films with different volume fractions of fillers as a function of the electric field summarized from DE loops. Reproduced with permission^[52]. Copyright 2017 American Chemical Society. c) Schematic models of the molecular arrangement of P(VDF-TrFE-CTFE) and PVDF-MS28. Reproduced with permission^[53]. Copyright 2019 Royal Society of Chemistry.

Modification of chemical structure with bulky monomers can help to form ferroelectric nanodomain and reduce the crystallinity, hence reducing the spontaneous polarization loss^[54]. Chu et al^[55] reported a kind of defect modified poly(vinylidene fluoride) polymers. These defects that expanded the interchain

lattice spacing might lower the energy of the TGTG' conformation and achieved a reversible change in conformations between the nonpolar and polar phases. Liu et al.^[53] reported a series of antiferroelectric-like poly (vinylidene fluoride-co-trifluoroethylene-co-chlorotrifluoroethylene)-graft-poly(styrene-methyl methacrylate) (PVDF-MS) copolymers synthesized via an atom transfer radical polymerization procedure (Fig. 5c). Antiferroelectric-like behaviors with relatively narrow hysteresis loops and low remnant polarization were achieved. Besides, irradiation has been demonstrated to have an effect on the charge-discharge process of ferroelectric dielectric films. Zhang et al.^[56] observed relaxor ferroelectric behavior in electron irradiated poly(vinylidene fluoride-trifluoroethylene) [P(VDF-TrFE)] copolymer, which suggested the electron irradiation broke up the coherent polarization domain (all-trans chains) in normal ferroelectric P(VDF-TrFE) copolymer into nanopolar regions and resulted in the slim polarization hysteresis loop.

In summary, optimizing of crystallization and orientation needs to be explored to improve the efficiency of ferroelectric polymers, as the efficiencies of most ferroelectric polymers are still unsatisfactory (less than 90%). Recently, Pan et al.^[57] found that substantial enhancements of energy storage properties could be achieved in relaxor ferroelectric ceramic films with a superparaelectric design. The nanodomains were scaled down to polar clusters of several unit cells, and polarization switching hysteresis was nearly eliminated while relatively high polarization was maintained. This can provide a reference for the development of polymer-based dielectrics. Moreover, capacitor operation is accompanied by obvious temperature rise. The energy loss of ferroelectric polymers at elevated temperatures also needs further investigations.

3.3 Polymer Blend

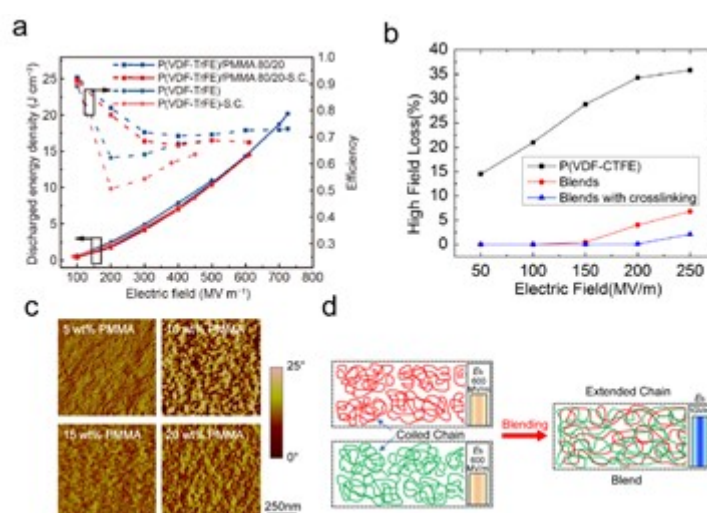


Figure 6 a) Variation of discharged energy density with electric field for pure P(VDF-TrFE), 80 vol% P(VDF-TrFE)/20 vol% PMMA blend films prepared by different methods. Reproduced with permission^[58]. Copyright

This article is protected by copyright. All rights reserved.

2021 Springer Nature. b) high field loss of pure P(VDF-CTFE) and blends films as a function of applied electric field. Reproduced with permission^[59]. Copyright 2011 American Institute of Physics. c) AFM images of blend composite with different weight fraction of PMMA. Reproduced with permission^[60]. Copyright 2019 Elsevier. d) Schematic illustration of the PI/PEI blend strategy in enhancing the electrostatic interaction and engineering the extended chain packing. Reproduced with permission^[61]. Copyright 2021 Elsevier.

Spontaneous polarization loss is the main source of loss during the charge-discharge process of ferroelectric polymers. Decreasing the loss of spontaneous polarization relies on decreasing the degree of polarization and increasing reversibility to reduce the ferroelectric switch. The introduction of linear dielectric is thought to be a potential method to suppress energy loss. The ferroelectric phase will endure a low electric field, as linear dielectrics with lower conductivity and lower permittivity endure a high electric field. Zhang et al^[58] demonstrated that polymer blends of poly(vinylidene fluoride-trifluoroethylene) (P(VDF-TrFE)) and polymethyl methacrylate (PMMA) showed tailored ferroelectric behaviors with double hysteresis loops and much suppressed remnant electrical displacement, resulting in improved efficiency (Fig. 6a). Wu et al^[59] developed a polar-fluoropolymer blend consisting of a high energy density poly(vinylidene fluoride-chlorotrifluoroethylene) (P(VDF-CTFE)) and a low dielectric loss poly(ethylene-chlorotrifluoroethylene) (ECTFE). Crosslinking was also introduced to minimize the loss from ferroelectric switch and conduction (Fig. 6b). In addition, Zhu et al^[60] found that the interaction between PMMA and poly(vinylidene fluoride-trifluoroethylene-chlorofluoroethylene) (P(VDF-TrFE-CFE)) polymer chains led to suppressed dielectric loss, small grain size and restricted leakage current density (Fig. 6c).

Recently, Feng et al^[62] prepared polymer blends consisting of P(VDF-HFP) and polyimide (PI). These composites showed much improved efficiency compared with pure PI at elevated temperatures, which was attributed to the suppressed space charge. Meanwhile, Zhang et al^[61] reported that strong interchain electrostatic interaction between two high-temperature polymers (PI with PEI) in blends yielded an extended polymer chain conformation, resulting in dense chain packing and improved charge-discharge efficiency (Fig. 6d). These two research results show that the performance of polymer blends is not just a compromise of properties of different phases. The effect of strong interaction of different phases on the conformation of polymers and charge dynamics may bring new challenges and opportunities in this field.

3.4 Polymer-filler Interface

Preparing polymer composites with functional filler has been widely adopted to enhance capacitive performances^[63]. However, as mentioned in the content of 2.2.2, the multiphase of composite dielectrics not only induces interfacial polarization loss, but also brings increased conduction loss and premature failure due to the local distortion of the electric field in the interphase. In addition, poor compatibility resulting from the difference in properties between different phases leads to weak insulation at the

interface, which further increases the energy loss. Therefore, the polymer-filler interface needs to be well designed in terms of the interaction between polymer and fillers to realize high-performance dielectric composites.

The coupling effect of the filler/matrix interface plays a prominent role in the dielectric properties of organic/inorganic nanocomposites^[7]. Core-shell structure is thought to be an effective way of suppressing the dielectric loss of polymer nanocomposites^[64]. Zheng et al^[65] coated ethylene propylene diene monomer (EPDM) as the shell on the surfaces of BaTiO₃ and added these nanoparticles into polypropylene (PP) (Fig. 7a). The dielectric loss of the EPDM@BT/PP composites was barely changed when compared with PP and improved tensile property was also achieved. Ren et al^[22b] reported the PEI composites filled with nanoparticles composed of ZrO₂ core and Al₂O₃ shell. The gradient of the dielectric constant from core to matrix reduced the electric field distortion around the nanoparticles. Also, the wide bandgap Al₂O₃ shell acted as deep traps in the composites. These composites showed an order of magnitude lower leakage of current density and much-improved efficiency with respect to those with pristine ZrO₂ at high temperatures (Fig. 7b). Zhang et al^[66] synthesized TiO₂ nanofibers embedded with BaTiO₃ nanoparticles and added them into the PVDF-HFP matrix. A high discharge efficiency of ~78% was achieved at an electric field of ~800 kV mm⁻¹. Zhang et al^[67] used three kinds of fluoro-polymers as the transition layer between the particles and the polymer matrix and found that the transition layer can effectively reduce the dielectric mismatch between fillers and matrix. The composite with the transition layer exhibited the discharge efficiency of 78% compared to 62% for a conventional solution blended composite.

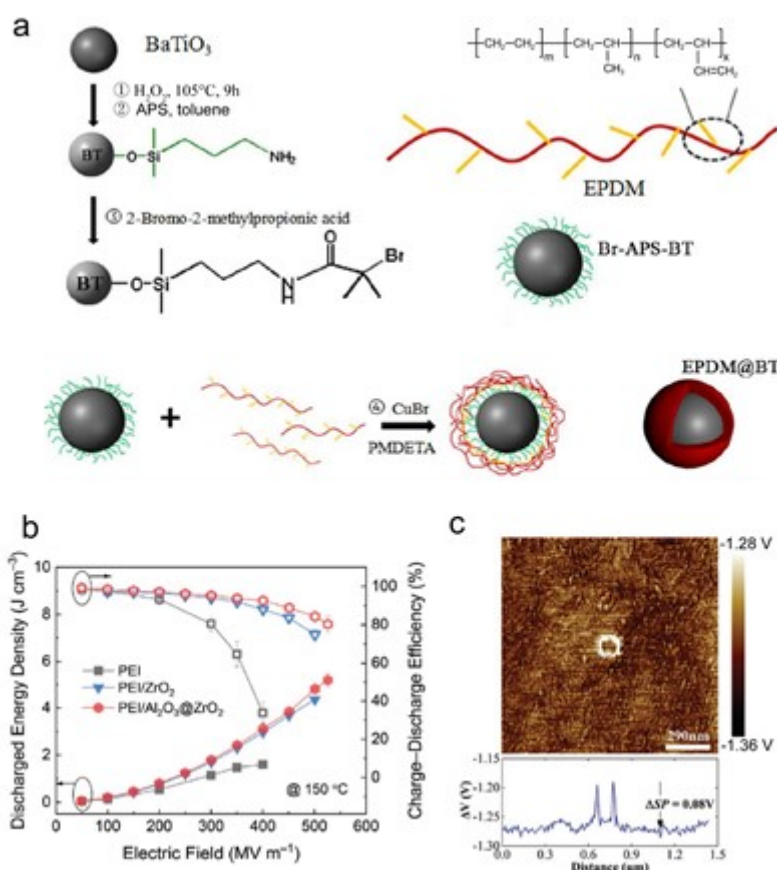


Figure 7 a) Schematic of synthetic process for core-shell structure EPDM@BT nanoparticles. Reproduced with permission^[65]. Copyright 2018 Elsevier. b) Discharged energy density and charge-discharge efficiency of PEI, the PEI nanocomposites with 11 vol% ZrO₂ and Al₂O₃@ZrO₂ nanoparticles as a function of electric field measured at 150 °C. Reproduced with permission^[22b]. Copyright 2021 John Wiley and Sons. c) The surface potential difference signal near an embedded nanoparticle Reproduced with permission^[68]. Copyright 2019 John Wiley and Sons.

Some progress has been made in the direct detection of local electric polarization in the interfacial region in ferroelectric polymer nanocomposites (Fig. 7c)^[68]. Yet, there is still a lack of direct characterizations of interface properties, such as the conductivity, breakdown strength and bandgap. These properties are essential for the loss analysis of nanocomposites. Therefore, the analysis of the role of the interface at high electric fields is still empirical. The in situ observation is essential for the understanding and development of polymer-filler interface design.

3.5 Utilization of Fillers Served as Deep Trap

Fillers used as deep traps show great potential for capturing carriers and reducing the conduction loss of polymer-based dielectric composites.

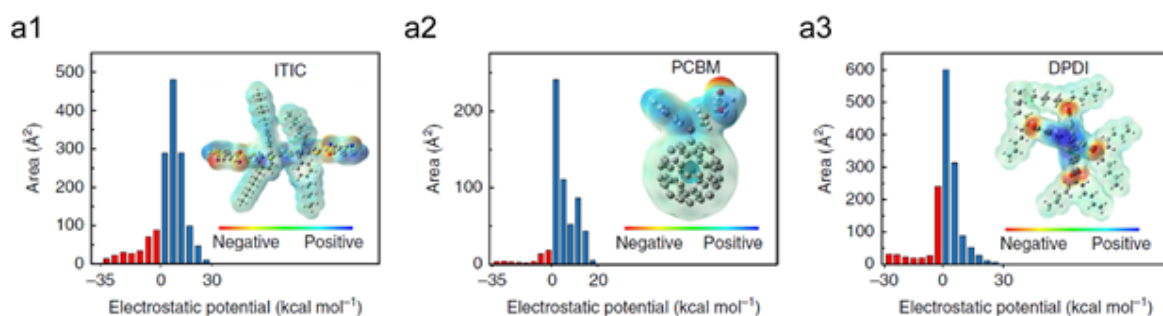


Figure. 8 Electrostatic potential distribution and area percentage in each electrostatic potential range of a1) ITIC (2,2'-[[6,6,12,12-Tetrakis(4-hexylphenyl)-6,12-dihydrodithieno[2,3-d:2',3'-d']-s-indaceno[1,2-b:5,6-b']dithiophene-2,8-diyl]bis[methyldiylidene(3-oxo-1H-indene-2,1(3H)-diylidene)]]bis[propanedinitrile]), a2) PCBM ([6,6]-Phenyl C61 butyric acid methyl ester), and a3) DPDI (2,2',9,9'-Tetrakis(1-pentylhexyl)-[5,5'-bianthra[2,1,9-def:6,5,10-d'e'f'] diisoquinoline]-1,1',3,3',8,8',10,10'(2H,2'H,9H,9'H)-octone)^[38]. Reproduced from ref. 38 with open access from Springer Nature.

Yuan et al.^[38] reported a series of all-organic polymer-based composites filled with molecular semiconductors (Fig. 8a). The molecular semiconductors immobilized free electrons via strong electrostatic attraction and impeded electric charge injection and transport in dielectric polymers, which led to concurrent high energy density (3.0 J cm^{-3}) and high discharge efficiency (90%) of the composite dielectrics (PCBM/PEI) up to 200°C . Wang et al.^[69] showed that hydrogen bonding network was formed between the glucose molecules and the polymer matrix. The hydrogen bonds could serve as trapping sites of charge carriers and network of hydrogen bonds could further facilitate increased crystallinity, reduced crystallite size, and stabilized γ phase in the polymer dielectrics, which led to enhanced electric displacement and reduced hysteresis loss. Thakur et al.^[70] found that semi-crystalline poly(tetrafluoroethylene-hexafluoropropylene-vinylidene fluoride) (P(TFE-HFP-VDF)) terpolymer with 1 wt.% (<0.5 vol. %) Al_2O_3 nanofillers showed more than two orders of magnitude reduction in conduction which was attributed to the decreased mobile carrier concentrations and increased trap depth.

Deep traps have been demonstrated to be effective in decreasing the conduction loss. Nevertheless, the role of traps was mainly inferred from the experimental result. There are few reports on the numerical description (models for calculation, simulations, et al.) of the effect of deep traps on polymer-based dielectrics, especially at high fields and elevated temperatures. In-depth understanding of the structure-formation-effect relationship of deep traps could drive the rational design of high-temperature polymer-based dielectrics.

3.6 Utilization of Wide Bandgap Fillers

Fillers with wide bandgap, especially 2-dimensional (2D) sheet fillers have received wide attention in this field. Li et al.^[71] reported crosslinked polymer nanocomposites with boron nitride nanosheets (BNNs)

(Fig. 9a). The electrical conductivity of these nanocomposites was several orders of magnitude lower than that of existing polymers. A superhierarchical nanocomposites of ferroelectric perovskites, layered aluminosilicate nanosheets, and an organic polymer matrix showed much improved efficiency compared with pristine polymer (Fig. 9b)^[72]. Also, Jiang et al^[73] prepared polymer nanocomposites with a novel interpenetrating gradient structure by cofilling PVDF with barium zirconate titanate (BZT) nanofibers and BNNS via modified nonequilibrium processing (Fig. 9c). The polymer nanocomposites showed suppressed electric conduction and enhanced breakdown strength. Besides, Dai et al^[74] fabricated nanocomposites consisting of polyimide-poly(amic acid) copolymer as matrix and BNNSs as fillers. The addition of BNNSs could further improve the charge-discharge efficiency of nanocomposites.

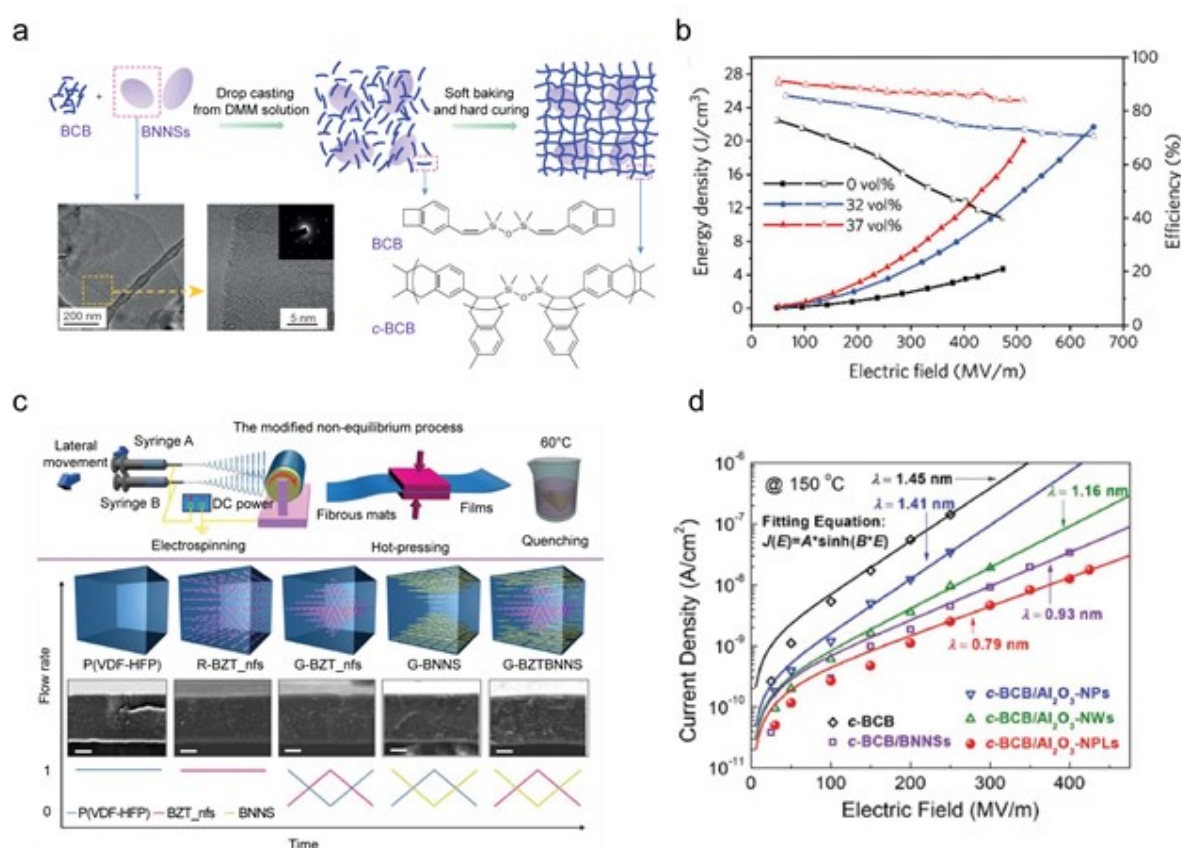


Figure. 9 a) Schematic of the preparation of c-BCB/BNNS films. Reproduced with permission^[71]. Copyright 2015 Springer Nature. b) Energy density and efficiency values of BT@AS/MMT/P(VDF-HFP) nanocomposite films under applied electric fields up to their breakdown fields. Reproduced with permission^[72]. Copyright 2020 John Wiley and Sons. c) Schematic illustration of the synthesis procedure of the gradient-structured polymer nanocomposites using the modified non-equilibrium process and cross-sectional SEM images (middle row, scale bar: 5 μ m) and the corresponding flow rate patterns (bottom row) of polymer nanocomposites with different configurations. Reproduced with permission^[73]. Copyright 2019 John Wiley and Sons. d) Leakage current density of c-BCB and the c-BCB nanocomposites as a function of electric field at 150 °C. Reproduced with permission^[22c]. Copyright 2019 John Wiley and Sons.

Notably, some kinds of fillers with wide bandgap could also function as deep traps, such as Al₂O₃.

This article is protected by copyright. All rights reserved.

However, it is believed that there exist two different mechanisms, which can be proved by the following two points. First, the conductive particles could act as deep traps and reduce the conductivity of polymer composite. Li et al^[75] prepared low-density polyethylene (LDPE)/graphene nanocomposites, which showed a lower DC conductivity, a higher breakdown strength and less space charge accumulation than neat LDPE. Second, the filler contents of composite dielectrics for the highest resistivity and breakdown strength are different for these two mechanisms. Wang's group^[22c] reported a solution-processable polymer nanocomposites consisting of readily prepared Al_2O_3 fillers with systematically varied morphologies including nanoparticles, nanowires and nanoplates (Fig. 9d). The optimal filler contents for the capacitive performance were 1.5 vol% for nanoparticles, 3.0 vol% for nanowires, and 7.5 vol% for nanoplates, respectively. The content of fillers mainly serving as deep traps were usually lower than 2.0 vol% and the optimum 2D sheet fillers with wide bandgap was usually around 10.0 vol%.

However, it should be noted that the bandgap and resistivity of fillers which are used for the analysis of composite performance are usually inferred from the results of bulky materials, while the fillers used in composite dielectrics are nanometer sized. The in-situ characterizations of nano-scale materials properties is essential for the understanding and development of polymer-based dielectrics with high bandgap fillers. Meanwhile, the distribution and orientation of fillers in the composite dielectrics have been demonstrated to have a great effect on the energy loss. Further regulations of the distribution and orientation as well as preparation optimization for mass production are important for the development of composite dielectrics.

3.7 Multilayer-Structure Dielectrics

Multilayer structures can be divided into two types according to whether the matrices of all layers are organic, namely organic multilayers and organic & inorganic multilayers.

Introducing linear dielectrics with low loss into ferroelectric polymer can regulate the electric field distribution and improve the efficiency. Baer et al^[76] prepared multilayer films, which contained up to 256 layers of alternating high permittivity (e.g., PVDF or relaxor ferroelectric terpolymers) and high breakdown/low loss polymers (e.g., polycarbonate(PC)) (Fig. 10a). The multilayer films could minimize ferroelectric switching in PVDF crystals and reduced dielectric loss from impurity ion hopping polarization. Chen et al^[77] selected PMMA as inner layer and PVDF-HFP as outer layer to prepare the trilayered all-polymer films, which showed much reduced dielectric loss. Huang et al^[78] prepared sandwiched multilayer films with polysulfone (PSF) as out layers and PVDF as inner layer. Unipolar electric field (DC + AC) poling at elevated temperatures was adopted to polarize impurity ions from the PVDF layers into the PSF layers, and impurity ions were largely "locked" in PSF (Fig. 10b) when the temperature dropped. The dissipation factor was as low as 0.003 at 1 Hz and 100°C. Besides, Niu et al^[79] found that sandwich-

This article is protected by copyright. All rights reserved.

structured polymer dielectrics (polyetherimides(PEIs)/PI/PEIs) showed improved efficiency compared with single layers of PI and PEIs at elevated temperatures. The interface barrier between different layers was thought to block the space charges and suppress the conduction loss.

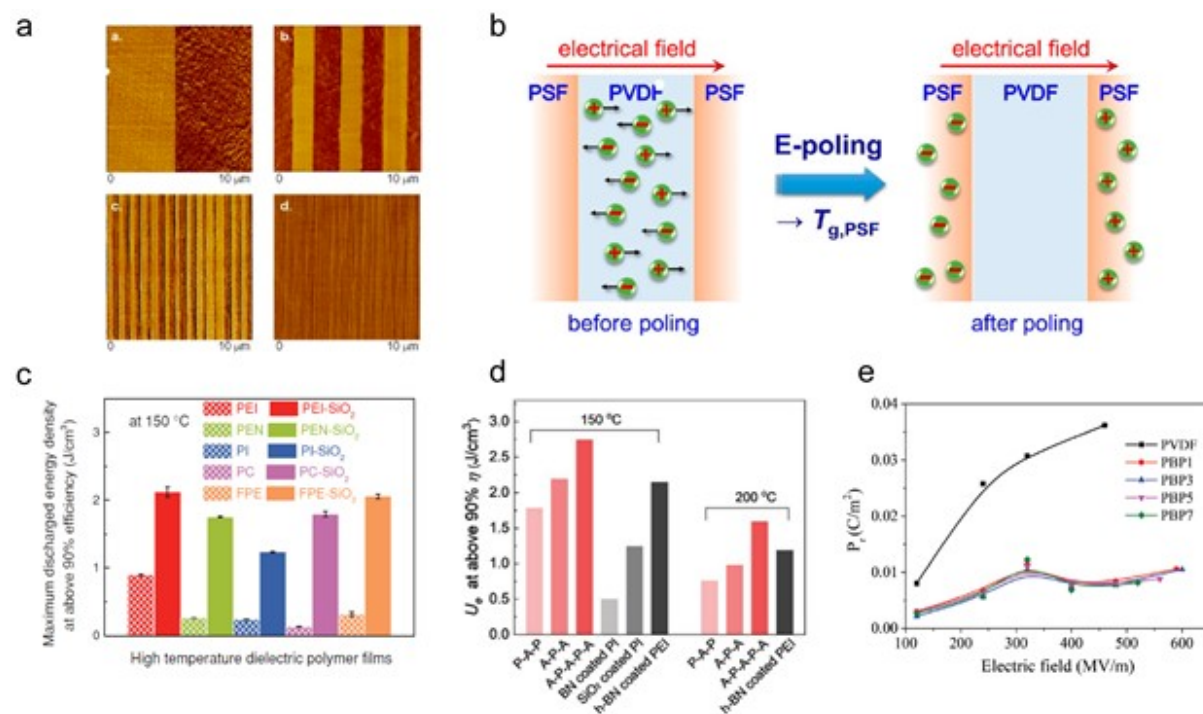


Figure. 10 a) Atomic Force Microscope (AFM) phase images of polycarbonate (PC)/PVDF 50/50 (vol/vol) multilayer films. Reproduced with permission^[76]. Copyright 2017 American Chemical Society. b) Schematic representation of electric poling near the glass transition temperature of PSF to drive impurity ions in the PVDF layer into neighboring PSF layers. Reproduced with permission^[78]. Copyright 2018 American Chemical Society. c) Maximum discharged energy density of the various dielectric films before and after coating achieved at above 90% charge–discharge efficiency measured at 150 °C. Reproduced with permission^[26b]. Copyright 2018 John Wiley and Sons. d) Comparison of the U_e at above 90% efficiency of P-A-P, A-P-A, A-P-A-P-A film (A represents Al_2O_3 and P represents PI), BN coated PI, h-BN coated PEI, and SiO_2 coated PI films at 150 and 200 °C. Reproduced with permission^[80]. Copyright 2021 John Wiley and Sons. e) Comparison of remnant polarization (P_r) of PVDF and sandwich-structured films with PVDF as outer layers and BNNs as middle layer (PBP) under various electric field (0.13, 0.16, 0.55, and 1.03 vol% of BNNs in PBP1, PBP3, PBP5, and PBP7, respectively). Reproduced with permission^[81]. Copyright 2019 John Wiley and Sons.

Also, the introduction of inorganic layer demonstrates its effectiveness in enhancing the efficiency. Azizi et al^[26a] reported that the improved performance of discharged energy density and charge–discharge efficiency at elevated temperatures was achieved in the PEI sandwiched with chemical vapor deposition(CVD)-grown *h*-BN films compared to neat PEI films. The *h*-BN layers effectively blocked the charge injection from electrodes and greatly reduces the dielectric loss. Zhou et al^[26b] proposed a plasma-enhanced chemical vapor deposition method to coat SiO_2 onto the surface of dielectric films. Introduced SiO_2 layer could increase the potential barrier at the electrode/dielectric interface, and consequently,

This article is protected by copyright. All rights reserved.

impeded the charge injection from electrodes and yielded much reduced space charge densities, which led to much improved charge-discharge efficiency (Fig. 10c). Dong et al.^[80] described a new approach based on the in-situ preparation of oxides onto PI films to high-temperature laminated polymer dielectrics, which showed remarkably improved efficiency. Charge injection at the electrode/dielectric interface and carrier transport within dielectrics were substantially depressed through engineering the oxide layer as charge-blocking layers (Fig. 10d). Zhu et al.^[81] prepared PVDF composite films with a nanosized interlayer of assembled BNNS that was aligned along the in-plane direction. The composite films showed the reduced remnant polarization (Fig. 10e). Zhang et al.^[82] proved that growing an inorganic insulation interlayer, such as MgO and BN, could significantly suppress the electrical conduction loss of PEI films at elevated temperature. Meanwhile, Wang et al.^[83] have shown that suppressed conduction and much enhanced charge-discharge efficiency were achieved in montmorillonite(MMT) coated polyamideimide (PAI) dielectrics. These inorganic layers with wide bandgap were believed to be effective in inhibition of charge injection and migration.

However, the properties of different layers in the multilayer composite vary greatly, which may lead to the redistribution of electric field and severe space charge accumulation. Furthermore, the injection and migration of space charge take a long time to fully play their role in the redistribution of electric field. Therefore, the long-term performance needs to be further investigated. Besides, the processing method should be carefully studied as it can affect the performance of each layer and the interaction between different phases in the dielectric composites. More attention needs to be paid to the adhesion between organic and inorganic phases. Physical vapor deposition, chemical vapor deposition and atomic layer deposition can be applied in the preparation of inorganic layers and need to be carefully compared and investigated for the further enhancement of efficiency.

4. Conclusions and outlook

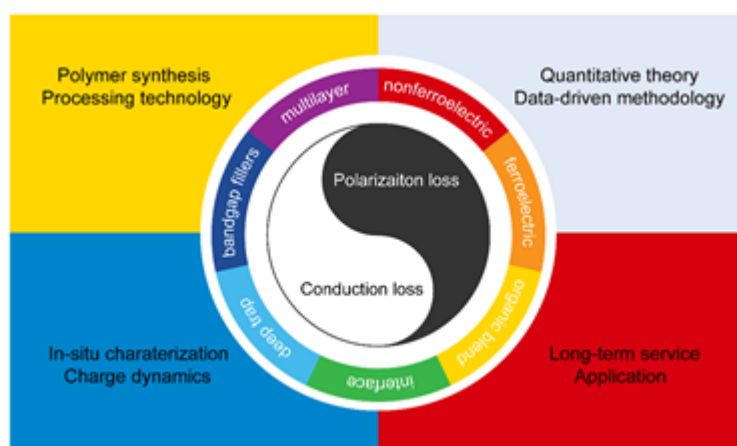


Figure 11 Schematic of research on the suppression of dielectric loss. The inner circle represents the two main

This article is protected by copyright. All rights reserved.

parts of the dielectric loss: polarization loss and conduction loss. The middle ring demonstrates the main strategies for dealing with the dielectric loss summarized in this review. The outer diagram depicts the existing challenges and prospects in this field.

This review has focused on the progress of loss suppression of polymer-based dielectrics for film capacitors. The dielectric loss is an essential factor to evaluate a material's potential in the application of capacitors. Research progress in understanding and suppressing the loss of polymer-based dielectrics has been summarized and discussed deeply in this review.

The dielectric loss is composed of polarization loss and conduction loss. For the polarization loss, the losses of dipolar polarization, spontaneous polarization and space charge polarization pose challenges and opportunities for the loss suppression of polymer-based dielectrics. As for the conduction loss, identifying the leakage current type is usually used to analyze the loss at high electric fields. For the loss measurement, in addition of characterizing the dielectric loss, the DE loop can provide more potential information in the charge-discharge process. For instance, the slope alteration in the DE loop can be used to analyze the change of charge and materials. However, the characterization and understanding of different loss types are far from satisfactory. For example, the classification of conduction loss is mainly based on the steady-state test. However, the DE loop is often measured at 10 Hz or 100 Hz and the leakage current cannot fully stabilize in the process. The dynamic of conduction and charge transportation is necessary for a deep understanding and analysis of conduction loss. Besides, reliable measurement of leakage current still requires particular attention, as the values of leakage current vary a lot across different studies. In addition, the effect of space charge polarization caused by charge injection from the electrode has been seldom discussed

From the view of material design, polymer-based dielectrics with different processing approaches, chemical structures, functional fillers and treatment processes have been reported. The suppression of polarization loss mainly deals with dipolar polarization and spontaneous polarization. Reducing the loss of the dipolar polarization mainly focuses on reducing the 'friction' of dipole orientation. Additionally, decreasing the loss of spontaneous polarization relies on decreasing the degree of polarization and increasing reversibility to reduce the ferroelectric switch. As for the conduction loss suppression, strategies can be divided into two categories: reducing carrier transport in the bulk and suppressing electrode-dielectric charge injection. Increasing bandgap, crosslinking and grafting of the polymer matrix can inhibit the charge transport and charge injection. Functional filler and polymer-filler interface design can help suppress the charge transport in the dielectrics. Selected nanointerlayers between the electrodes and dielectrics can reduce the charge injection from the electrodes. Considering the advantages of loss suppression and the possibility of mass production, the design of linear polymer dielectrics, the utilization of deep traps and nanointerlayers between electrodes and dielectrics may be

This article is protected by copyright. All rights reserved.

promising candidates for the design of dielectrics for capacitor applications.

Substantial progress has been made in the research of loss suppression of polymer-based dielectrics. Yet, there still exist some scientific and technical challenges. Further devotion to this field may include the following four aspects for the development of polymer-based dielectric with high efficiency:

1) Polymer synthesis and processing technology.

It is still unclear whether the spontaneous polarization can be applied in polymer-based dielectrics for capacitors. The loss induced by dipole polarization is worthy of a more in-depth investigation, as the dipole polarization can enhance the energy density of dielectrics and has a much less negative effect on the efficiency compared with spontaneous polarization. Therefore, the design of the chemical structure and processing technology deserves systematic investigations through continuous efforts.

2) In-situ characterization and charge dynamics.

The measurements of structure, chemical and electrical properties of the fillers and interface are vital for the analysis of dielectric performances. The mechanisms of enhanced energy density and suppressed loss of polymer composites with ultra-low loading of fillers need further investigations as the interaction between fillers and polymer matrix is still unclear. due to the lack of in-situ characterizations. The effect of space charge injected from the electrodes demands further investigations especially at high fields and elevated temperatures. Characterization techniques and numerical simulations of the charge injection and migration in the dielectrics can facilitate the design of advanced dielectric polymer materials.

3) Quantitative theory and data-driven methodology

Many theories or mechanisms in the suppression of energy loss seem to be empirical and lack numerical description (models for calculation, simulations et al). For instance, there is no clear quantitative relationship between the parameters of the deep trap and the leakage current. The computation and data-driven methods have shown their advantages in the design of polymer structures with wide bandgap and low loss. Prediction of structure and performance before material preparation can greatly increase efficiency and accelerate the development of polymer-based materials.

4) Long-term service and application

Few studies have been reported on long-term service performances of film capacitors. Functional fillers with wide bandgap or serving as deep traps have demonstrated their roles in the suppression of energy loss. Meanwhile, the multilayer design shows its advantages in improving efficiency, especially for

This article is protected by copyright. All rights reserved.

polymer films covered with inorganic nanolayers. However, these strategies may also be accompanied with issues such as the redistribution of space charge and electric field within the materials, which may lead to accelerated aging of materials and premature failure of capacitors. In addition, the high loading of inorganic fillers may lead to deteriorated flexibility, which is unfavorable for the mass production of film capacitors. Some polymer-based dielectrics lack self-healing capabilities. Nevertheless, the applications of polymer-based dielectrics in capacitors require favorable processability and sufficient self-clearing performances. These performances need considerable research efforts in the research of loss suppression.

In general, the suppression of loss appeals to collaborative efforts from chemistry, materials science and electrical engineering. With the help of data-driven and numerical simulations, advanced material preparation techniques and characterization methods, the development of polymer-based dielectrics will achieve critical advances.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (No. 51937007 and 51921005) and National Key Research and Development Program of China (No. 2021YFB2401502).

Reference

- [1] T. M. Gür, *Energy Environ. Sci.* **2018**, 11, 2696.
- [2] a) A. S. Arico, P. Bruce, B. Scrosati, J. M. Tarascon, W. van Schalkwijk, *Nat. Mater.* **2005**, 4, 366; b) H. S. Chen, T. N. Cong, W. Yang, C. Q. Tan, Y. L. Li, Y. L. Ding, *Prog. Nat. Sci.: Mater. Int.* **2009**, 19, 291.
- [3] a) Z. M. Dang, J. K. Yuan, S. H. Yao, R. J. Liao, *Adv. Mater.* **2013**, 25, 6334; b) A. R. Dehghani-Sanij, E. Tharumalingam, M. B. Dusseault, R. Fraser, *Renewable Sustainable Energy Rev.* **2019**, 104, 192; c) S. L. Zhong, Z. M. Dang, W. Y. Zhou, H. W. Cai, *IET Nanodielectrics* **2018**, 1, 41.
- [4] a) Z. M. Dang, J. K. Yuan, J. W. Zha, T. Zhou, S. T. Li, G. H. Hu, *Prog. Mater. Sci.* **2012**, 57, 660; b) H. X. Chen, Z. B. Pan, W. L. Wang, Y. Y. Chen, S. Xing, Y. Cheng, X. P. Ding, J. J. Liu, J. W. Zhai, J. H. Yu, *Composites. A* **2021**, 142, 106266; c) H. Li, Y. Zhou, Y. Liu, L. Li, Y. Liu, Q. Wang, *Chem. Soc. Rev.* **2021**, 50, 6369; d) Z. M. Dang, *Dielectric polymer materials for high-density energy storage*, William Andrew, **2018**.
- [5] T. D. Huan, S. Boggs, G. Teyssedre, C. Laurent, M. Cakmak, S. Kumar, R. Ramprasad, *Prog. Mater. Sci.* **2016**, 83, 236.
- [6] H. L. Hu, F. Zhang, S. B. Luo, W. K. Chang, J. L. Yue, C. H. Wang, *Nano Energy* **2020**, 74, 104844.
- [7] Z. Pan, L. Yao, J. Zhai, X. Yao, H. Chen, *Adv. Mater.* **2018**, 30, e1705662.
- [8] a) M. Streibl, R. Karmazin, R. Moos, *IEEE Trans. Dielectr. Electr. Insul.* **2018**, 25, 2429; b) L. Zhu, *J. Phys. Chem. Lett.* **2014**, 5, 3677.
- [9] a) L. Zhu, Q. Wang, *Macromolecules* **2012**, 45, 2937. b) M. Yang, Z. Y. Xue, Z. M. Dang, Y. Shen, P. Haghi-Ashtiani, D. He, J. Xu, J. Bai, *IET Nanodielectrics* **2021**, 4, 11.
- [10] Y. Zhou, Q. Wang, *J. Appl. Phys.* **2020**, 127, 240902.
- [11] Q. Li, F. Z. Yao, Y. Liu, G. Zhang, H. Wang, Q. Wang, *Annual Rev. Mater. Res.*, **2018**, 48, 219 (Ed: D. R. Clarke).
- [12] H. Li, L. L. Ren, Y. Zhou, B. Yao, Q. Wang, *High Voltage* **2020**, 5, 365.
- [13] K. C. Kao, in *Dielectric Phenomena in Solids*, (Ed: K. C. Kao), Academic Press, San Diego **2004**.
- [14] D. A. Seanor, in *Electrical Properties of Polymers*, (Ed: D. A. Seanor), Academic Press, **1982**.
- [15] a) F. C. Chiu, *Adv. Mater. Sci. Eng.* **2014**, 2014, 1; b) M. Ieda, *IEEE Trans. Electr. Insul.* **1984**, 19, 162; c) B. Hadri, P. R. Mamy, J. Martinez, M. Mostefa, *Solid State Commun.* **2006**, 139, 35; d) M. Ieda, M. Nagao, M. Hikita, *IEEE Trans. Electr. Insul.* **1994**, 1, 934.
- [16] S. Barth, U. Wolf, H. Bassler, P. Muller, H. Riel, H. Vestweber, P. F. Seidler, W. Riess, *Phys. Rev. B* **1999**, 60, 8791.
- [17] G. Lengyel, *J. Appl. Phys.* **1966**, 37, 807.
- [18] S. Zaima, T. Furuta, Y. Koide, Y. Yasuda, M. Iida, *ChemInform* **1990**, 21.
- [19] Q. Chen, Y. Wang, X. Zhou, Q. M. Zhang, S. H. Zhang, *Appl. Phys. Lett.* **2008**, 92, 142909.

This article is protected by copyright. All rights reserved.

- [20] V. Ambegaokar, B. I. Halperin, J. S. Langer, *Phys. Rev. B.* **1971**, 4, 2612.
- [21] J. N. Ho, T. R. Jow, *IEEE Trans. Electr. Insul.* **2012**, 19, 990.
- [22] a) D. Ai, H. Li, Y. Zhou, L. Ren, Z. Han, B. Yao, W. Zhou, L. Zhao, J. Xu, Q. Wang, *Adv. Energy Mater.* **2020**, 10, 1903881; b) L. Ren, H. Li, Z. Xie, D. Ai, Y. Zhou, Y. Liu, S. Zhang, L. Yang, X. Zhao, Z. Peng, R. Liao, Q. Wang, *Adv. Energy Mater.* **2021**, 11, 2101297; c) H. Li, D. Ai, L. Ren, B. Yao, Z. Han, Z. Shen, J. Wang, L. Q. Chen, Q. Wang, *Adv. Mater.* **2019**, 31, e1900875.
- [23] A. J. Campbell, D. D. C. Bradley, D. G. Lidzey, *J. Appl. Phys.* **1997**, 82, 6326.
- [24] G. C. Montanari, P. H. F. Morshuis, *IEEE Trans. Electr. Insul.* **2005**, 12, 754.
- [25] a) J. Y. Pei, J. W. Zha, W. Y. Zhou, S. J. Wang, S. L. Zhong, L. J. Yin, M. S. Zheng, H. W. Cai, Z. M. Dang, *Appl. Phys. Lett.* **2019**, 114, 103702; b) B. Zhang, J. Liu, M. Ren, C. Wu, T. J. Moran, S. Zeng, S. E. Chavez, Z. Hou, Z. Li, A. M. LaChance, T. R. Jow, B. D. Huey, Y. Cao, L. Sun, *Adv. Mater.* **2021**, e2101374; c) S. B. Luo, T. Q. Ansari, J. Y. Yu, S. H. Yu, P. P. Xu, L. Q. Cao, H. T. Huang, R. Sun, *Chem. Eng. J.* **2021**, 412, 128476; d) S. Cheng, Y. Zhou, Y. S. Li, C. Yuan, M. C. Yang, J. Fu, J. Hu, J. L. He, Q. Li, *Energy Storage Mater.* **2021**, 42, 445. e) D. Q. Tan, *IET Nanodielectrics* **2020**, 3, 28.
- [26] a) A. Azizi, M. R. Gadinski, Q. Li, M. A. AlSaud, J. Wang, Y. Wang, B. Wang, F. Liu, L. Q. Chen, N. Alem, Q. Wang, *Adv. Mater.* **2017**, 29; b) Y. Zhou, Q. Li, B. Dang, Y. Yang, T. Shao, H. Li, J. Hu, R. Zeng, J. He, Q. Wang, *Adv. Mater.* **2018**, 30, e1805672; c) J. Y. Pei, S. L. Zhong, Y. Zhao, L. J. Yin, Q. K. Feng, L. Huang, D. F. Liu, Y. X. Zhang, Z. M. Dang, *Energy Environ. Sci.* **2021**, 14, 5513.
- [27] a) S. Park, C. H. Kim, W. J. Lee, S. Sung, M. H. Yoon, *Mater. Sci. Eng., R* **2017**, 114, 1; b) Prateek, V. K. Thakur, R. K. Gupta, *Chem. Rev.* **2016**, 116, 4260.
- [28] M. Zhang, B. Li, J. J. Wang, H. B. Huang, L. Zhang, L. Q. Chen, *Adv. Mater.* **2021**, e2008198.
- [29] Z. Zhang, J. Zheng, K. Premasiri, M. H. Kwok, Q. Li, R. Li, S. Zhang, M. H. Litt, X. P. A. Gao, L. Zhu, *Mater. Horiz.* **2020**, 7, 592.
- [30] Y. Zheng, K. Chonung, G. Wang, P. Wei, P. Jiang, *J. Appl. Polym. Sci.* **2009**, 111(2), 917..
- [31] a) N. A. Nikonorova, A. A. Kononov, H. T. Dao, R. A. Castro, *J. Non-Cryst. Solids* **2019**, 511, 109; b) J. T. Bandler, D. A. Boyles, C. A. Edmondson, T. Filipova, J. J. Fontanella, M. A. Westgate, M. C. Wintersgill, *Macromolecules* **2013**, 46, 4024.
- [32] J. Wei, Z. Zhang, J. K. Tseng, I. Treufeld, X. Liu, M. H. Litt, L. Zhu, *ACS Appl. Mater. Interfaces* **2015**, 7, 5248.
- [33] a) L. Yang, X. Li, E. Allahyarov, P. L. Taylor, Q. M. Zhang, L. Zhu, *Polymer* **2013**, 54, 1709; b) L. Yang, J. Ho, E. Allahyarov, R. Mu, L. Zhu, *ACS Appl. Mater. Interfaces* **2015**, 7, 19894; c) M. R. Gadinski, Q. Li, G. Zhang, X. Zhang, Q. Wang, *Macromolecules* **2015**, 48, 2731; d) M. Yuan, B. Li, S. Zhang, R. Rajagopalan, M. T. Lanagan, *ACS Appl. Polym. Mater.* **2020**, 2, 1356.
- [34] S. L. Zhong, Z. M. Dang, J. W. Zha, *IEEE Trans. Electr. Insul.* **2018**, 25, 2122.
- [35] a) D. W. Davidson, R. H. Cole, *J. Chem. Phys.* **1951**, 19, 1484; b) L. A. Dissado, R. M. Hill, *J. Chem. Soc., Faraday Trans. II* **1984**, 80, 291; c) S. Havriliak, S. Negami, *Polymer* **1967**, 8, 161.

- [36] Q. Wang, X. Liang, C. Wu, N. Wang, S. Liu, Z. Zuo, Y. Gao, *Polymer* **2021**, 229, 124016.
- [37] Y. Jiang, X. Zhang, Z. H. Shen, X. H. Li, J. J. Yan, B. W. Li, C. W. Nan, *Adv. Funct. Mater.* **2020**, 30, 1906112.
- [38] C. Yuan, Y. Zhou, Y. Zhu, J. Liang, S. Wang, S. Peng, Y. Li, S. Cheng, M. Yang, J. Hu, B. Zhang, R. Zeng, J. He, Q. Li, *Nat. Commun.* **2020**, 11.
- [39] D. F. Miranda, S. H. Zhang, J. Runt, *Macromolecules* **2017**, 50, 8083.
- [40] H. Li, M. R. Gadinski, Y. Huang, L. Ren, Y. Zhou, D. Ai, Z. Han, B. Yao, Q. Wang, *Energy Environ. Sci.* **2020**, 13, 1279.
- [41] T. Zhang, X. Chen, Y. Thakur, B. Lu, Q. Zhang, J. Runt, Q. M. Zhang, *Sci. Adv.* **2020**, 6, eaax6622.
- [42] C. Wu, A. A. Deshmukh, Z. Li, L. Chen, A. Alamri, Y. Wang, R. Ramprasad, G. A. Sotzing, Y. Cao, *Adv. Mater.* **2020**, 32, e2000499.
- [43] Y. Tang, W. Xu, S. Niu, Z. Zhang, Y. Zhang, Z. Jiang, *J. Mater. Chem. A* **2021**, 9, 10000.
- [44] H. Yuan, Y. Zhou, Y. Zhu, S. Hu, C. Yuan, W. Song, Q. Shao, Q. Zhang, J. Hu, Q. Li, J. He, *J. Phys. D.* **2020**, 53, 475301.
- [45] Y. Zhang, Z. Liu, L. Zhu, J. Liu, Y. Zhang, Z. Jiang, *CCS Chemistry* **2020**, 2, 1169.
- [46] Y. Thakur, T. Zhang, C. Iacob, T. Yang, J. Bernholc, L. Q. Chen, J. Runt, Q. M. Zhang, *Nanoscale* **2017**, 9, 10992.
- [47] V. Sharma, C. Wang, R. G. Lorenzini, R. Ma, Q. Zhu, D. W. Sinkovits, G. Pilania, A. R. Oganov, S. Kumar, G. A. Sotzing, S. A. Boggs, R. Ramprasad, *Nat. Commun.* **2014**, 5, 4845.
- [48] Q. K. Feng, S. L. Zhong, J. Y. Pei, Y. Zhao, D. L. Zhang, D. F. Liu, Y. X. Zhang, Z. M. Dang, *Chem. Rev.* **2022**, 122, 3820.
- [49] a) X. T. Ren, N. Meng, H. Zhang, J. Y. Wu, I. Abrahams, H. X. Yan, E. Bilotti, M. J. Reece, *Nano Energy* **2020**, 72, 104662; b) J. J. Li, Q. J. Meng, W. J. Li, Z. C. Zhang, *J. Appl. Polym. Sci.* **2011**, 122, 1659; c) M. R. Gadinski, K. Han, Q. Li, G. Zhang, W. Reainthippayasakul, Q. Wang, *ACS Appl. Mater. Interfaces* **2014**, 6, 18981.
- [50] N. Meng, X. Ren, G. Santagiuliana, L. Ventura, H. Zhang, J. Wu, H. Yan, M. J. Reece, E. Bilotti, *Nat. Commun.* **2019**, 10, 4535.
- [51] R. Guo, H. Luo, X. F. Zhou, Z. D. Xiao, H. R. Xie, Y. Liu, K. C. Zhou, Z. H. Shen, L. Q. Chen, D. Zhang, *J. Mater. Chem. A* **2021**, 9, 27660.
- [52] J. Jiang, X. Zhang, Z. Dan, J. Ma, Y. Lin, M. Li, C. W. Nan, Y. Shen, *ACS Appl. Mater. Interfaces* **2017**, 9, 29717.
- [53] J. J. Liu, J. N. Liao, Y. Liao, Z. C. Zhang, *Polymer Chemistry* **2019**, 10, 3547.
- [54] a) B. Chu, X. Zhou, K. Ren, B. Neese, M. Lin, Q. Wang, F. Bauer, Q. M. Zhang, *Science* **2006**, 313, 334; b) J. Li, S. Tan, S. Ding, H. Li, L. Yang, Z. Zhang, *J. Mater. Chem.* **2012**, 22, 23468; c) R. Su, J. K. Tseng, M. S. Lu, M. Lin, Q. Fu, L. Zhu, *Polymer* **2012**, 53, 728.

- [55] A. J. Lovinger, *Science* **1983**, 220, 1115.
- [56] Q. M. Zhang, V. V. Bharti, X. Zhao, *Science* **1998**, 280, 2101.
- [57] H. Pan, S. Lan, S. Xu, Q. Zhang, H. Yao, Y. Liu, F. Meng, E. J. Guo, L. Gu, D. Yi, X. Renshaw Wang, H. Huang, J. L. MacManus-Driscoll, L. Q. Chen, K. J. Jin, C. W. Nan, Y. H. Lin, *Science* **2021**, 374, 100.
- [58] X. Zhang, Y. Jiang, R. Gao, X. Li, Z. Shen, B. W. Li, Q. Zhang, S. Zhang, C. W. Nan, *Sci. China Mater.* **2021**, 64, 1642.
- [59] S. Wu, M. Lin, S. G. Lu, L. Zhu, Q. M. Zhang, *Appl. Phys. Lett.* **2011**, 99, 132901.
- [60] Y. Zhu, P. Jiang, X. Huang, *Compos. Sci. Technol.* **2019**, 179, 115.
- [61] Q. Zhang, X. Chen, B. Zhang, T. Zhang, W. Lu, Z. Chen, Z. Liu, S. H. Kim, B. Donovan, R. J. Warzoha, E. D. Gomez, J. Bernholc, Q. M. Zhang, *Matter* **2021**, 4, 2448.
- [62] Q. K. Feng, D. F. Liu, Y. X. Zhang, J. Y. Pei, S. L. Zhong, H. Y. Hu, X. J. Wang, Z. M. Dang, *Nano Energy* **2022**, 99, 107410.
- [63] W. Zhou, T. Li, M. Yuan, B. Li, S. Zhong, Z. Li, X. Liu, J. Zhou, Y. Wang, H. Cai, *Energy Storage Mater.* **2021**, 42, 1.
- [64] a) X. Huang, P. Jiang, *Adv. Mater.* **2015**, 27, 546; b) B. Liu, M. H. Yang, W. Y. Zhou, H. W. Cai, S. L. Zhong, M. S. Zheng, Z. M. Dang, *Energy Storage Mater.* **2020**, 27, 443.
- [65] M. S. Zheng, Y. T. Zheng, J. W. Zha, Y. Yang, P. Han, Y. Q. Wen, Z. M. Dang, *Nano Energy* **2018**, 48, 144.
- [66] X. Zhang, Y. Shen, B. Xu, Q. Zhang, L. Gu, J. Jiang, J. Ma, Y. Lin, C. W. Nan, *Adv. Mater.* **2016**, 28, 2055.
- [67] T. Zhang, M. Guo, J. Jiang, X. Zhang, Y. Lin, C. W. Nan, Y. Shen, *RSC Advances* **2019**, 9, 35990.
- [68] S. Peng, X. Yang, Y. Yang, S. Wang, Y. Zhou, J. Hu, Q. Li, J. He, *Adv. Mater.* **2019**, 31, e1807722.
- [69] R. Wang, H. Xu, S. Cheng, J. Liang, B. Gou, J. Zhou, J. Fu, C. Xie, J. He, Q. Li, *Energy Storage Mater.* **2022**.
- [70] Y. Thakur, M. H. Lean, Q. M. Zhang, *Appl. Phys. Lett.* **2017**, 110, 122905.
- [71] Q. Li, L. Chen, M. R. Gadinski, S. Zhang, G. Zhang, U. Li, E. Iagodkine, A. Haque, L. Q. Chen, N. Jackson, Q. Wang, *Nature* **2015**, 523, 576.
- [72] B. C. Luo, Z. H. Shen, Z. M. Cai, E. K. Tian, Y. Yao, B. W. Li, A. Kursumovic, J. L. MacManus-Driscoll, L. T. Li, L. Q. Chen, X. H. Wang, *Adv. Funct. Mater.* **2021**, 31, 2007994.
- [73] J. Y. Jiang, Z. H. Shen, X. K. Cai, J. F. Qian, Z. K. Dan, Y. H. Lin, B. L. Liu, C. W. Nan, L. Q. Chen, Y. Shen, *Adv. Energy Mater.* **2019**, 9, 1803411.
- [74] Z. Dai, Z. Bao, S. Ding, C. Liu, H. Sun, H. Wang, X. Zhou, Y. Wang, Y. Yin, X. Li, *Adv. Mater.* **2022**, 34, e2101976.
- [75] Z. Li, B. Du, C. Han, H. Xu, *Sci Rep* **2017**, 7, 4015.
- [76] E. Baer, L. Zhu, *Macromolecules* **2017**, 50, 2239.

- [77] J. Chen, Y. Wang, Q. Yuan, X. Xu, Y. Niu, Q. Wang, H. Wang, *Nano Energy* **2018**, 54, 288.
- [78] H. D. Huang, X. Y. Chen, K. Z. Yin, I. Treufeld, D. E. Schuele, M. Ponting, D. Langhe, E. Baer, L. Zhu, *ACS Appl. Energy Mater.* **2018**, 1, 775.
- [79] Y. Niu, J. Dong, Y. He, X. Xu, S. Li, K. Wu, Q. Wang, H. Wang, *Nano Energy* **2022**, 97, 107215.
- [80] J. Dong, R. Hu, X. Xu, J. Chen, Y. Niu, F. Wang, J. Hao, K. Wu, Q. Wang, H. Wang, *Adv. Funct. Mater.* **2021**, 31, 2102644.
- [81] Y. Zhu, Y. Zhu, X. Huang, J. Chen, Q. Li, J. He, P. Jiang, *Adv. Energy Mater.* **2019**, 1901826.
- [82] T. Zhang, L. Yang, C. Zhang, Y. Feng, J. Wang, Z. Shen, Q. Chen, Q. Lei, Q. Chi, *Mater. Horiz.* **2022**.
- [83] Y. Wang, Z. Li, C. Wu, P. Zhou, J. Zhou, J. Huo, K. Davis, A. C. Konstantinou, H. Nguyen, Y. Cao, *Chem. Eng. J.* **2022**, 437, 135430.

Author's Biographies

Jia-Yao Pei received his B.S. degree from Tsinghua University, Beijing China, in 2017. Currently, he is working toward his Ph.D. degree under the supervision of Professor Zhi-Min Dang in the Department of Electrical Engineering, Tsinghua University. His research interests include polymer-based dielectrics, electrode-dielectric interface engineering and dielectric phenomena at high electric fields.



Zhi-Min Dang received his Ph. D. degree in electrical engineering from Xi'an Jiaotong University in 2001. He is currently a Professor in the Department of Electrical Engineering, Tsinghua University. His research interests are Advanced Energy/Electrical Materials and Devices. He has published more than 350 journal papers with a current citation record of over 16000 times and an H-index of 66.

How to suppress the loss of polymer-based dielectrics has been increasingly recognized as a critical point to attain a high charge-discharge efficiency in the film capacitors. The theoretical progress, measurement of properties, and development of materials are reviewed in this article. The challenges and opportunities in loss suppression research for the rational design of high-efficiency polymer-based dielectrics are proposed.



This article is protected by copyright. All rights reserved.